

Problem Statement & Client Needs:

Construct a rover capable of responding to mass casualty incidents (MCIs) by administering basic first aid before first responders are present on the scene.

Client Needs: The rover must be able to store medical equipment, navigate terrain, and provide limited medical assistance to victims. Clients would likely be first responders such as firefighters, who can use the rover to do an initial sweep of a scene before sending in people.

Objectives:

Primary: To construct an operational rover capable of providing basic first aid to victims in MCIs by utilizing both mechanical and electronic additions to the rover base.

Additional Functions: Many additions will be added to the rover to achieve these goals. This includes a servo driven robotic arm made entirely of 3D print material, a 3D printed fluid container with a built-in pump, and a 3D printed storage bin located on the front of the rover for storing medical supplies such as tourniquets to name a few.

Product Design Specifications

Functions: The rover will consist of many mechanical add-ons such as the arm, fluid tank, and storage bins to provide limited medical care to those in need.

Materials: The arm, fluid tank, and storage bin will be made of PLA. They will also utilize supporting electronics (servos, etc.)

Fabrication: All parts will be 3D printed using PLA, and the rover will be assembled by hand.

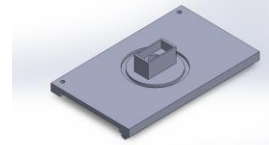
Equipment: Metal frame, wires, servo motors, breadboards, 3D printer, circuit boards, tubes, pump.

Notable Components

Custom Printed

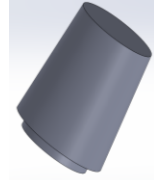
Base Plate:

The F.A.R.R. features a custom 3D printed baseplate used as the base for the servo driven arm. It features a rail slot on one side so that it may slide into the rover's built-in rails with screw holes on the other side to secure it to the frame. Additionally, there is a mount for a servo in the center for the base of the arm to rotate, as well as a track to help guide the rotation of the base of the arm. (Figure 1.1)



Rotary Arm Link:

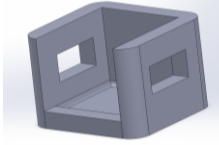
The arm link is seated within the baseplate track to provide guided rotational motion driven by the baseplate's mounted servo. It transfers the servo-driven torque into left-right rotation of the robotic arm while maintaining alignment and structural support. (Figure 1.2)



Dual Servo Mounting

Bracket:

The function of this component is to hold and secure two operative servos that lift and lower our robotic arm. It was then mounted upon the rotating arm link via screws through drilled mounting holes. (Figure 1.3)



Servo-Actuated Claw:

The claw was designed as the robotic arm's end effector, enabling the manipulation of objects during operation. Rotational input is transferred through the gear train, allowing controlled opening and closing via the linked potentiometer. The gripper was built from 3D-printed components, including gears, linkage arms, claw fingers, connector studs, and a mounting frame that supports the motor and main structure. (Figure 1.4)

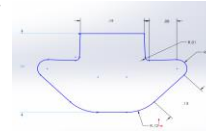


Design Obstacles:

Our first design concern centered around how we would mount our additional functionalities onto the given rover base. We landed on taking advantage of the modular nature of the rails that made up the rover frame. A CAD sketch was created of the rail negative (Fig. 1.5) and utilized to act as a modular mounting system which allowed for ease of repair and cleaning. Our other foremost challenging design challenges was our method of giving our servo arm rotational capabilities. Our solution was to vertically mount a singular servo on the interior of our track guided rotary arm link (Fig 1.2). This gave us precision to control the direction of our assisting arm and claw.

Figure 1.5

The negative of the rover Frame rail.



Timeline

Brainstorming → SolidWorks Design → 3D Printing → Assemble Parts → Electrical Components → Testing → Failure Review → Final Improvements

Conclusions

The F.A.R.R. project demonstrated how a rover can be modified with 3D-printed components to support first aid in mass-casualty situations. Although the final design had issues with arm stability, servo support, and overall weight distribution, these failures help us identify the main areas that would need improvement in a future version. Strengthening the base support, reducing unnecessary weight, and tweaking the arm connections all will allow for the F.A.R.R. better equipped to save more lives.

Acknowledgements:

Thank you to Dr. Kranov and the Engineering Department for their constant support throughout the process of this project.

Areas of Failure:

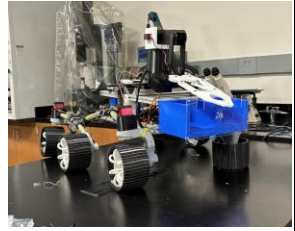
As seen in the video, our main area of failure and compromise is our servo-operated arm. Overall, there is an error in our base construction. We did not account for the front heavy nature of the arm and servo claw the revolving base had to support. The second main domain of error was arm linkages themselves. The pin tolerances were too great and lead to weak connecting joints which ultimately destabilized overall integrity.

Impact and Current Solutions:

There lacks any true land-based alternatives that assist first responders as the F.A.R.R. intends to do so. UAVs are currently used to help visually but also aid in the delivery of vital medical supplies into dangerous terrain ahead of human responders. With the current implementation of A.I.-based software systems, these drones have become increasingly effective. Whether due to material constraints or decreased degree of potency, land-based drones such as the F.A.R.R. are rarely implemented.

Figure 1.6

The F.A.R.R. after construction and wiring was completed.



QR Code:

Scan for access to MRAF demonstration video!

